Roots of d-matching polynomials and the spectrum of the universal cover: a closed form at first Betti number one, and a refuted conjecture

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Abstract

Hall, Puder and Sawin place the roots of the d-matching polynomial $\mu_{d,G}$ in the interval $[-\rho, \rho]$, ρ the spectral radius of the universal cover T. We conjectured that they lie in $\mathrm{spec}(T)$ itself, a proper subset of that interval whenever T has a spectral gap: no root ever enters a gap. This strengthens their theorem and is the matching-polynomial form of their Question 6.3.

We prove it for every connected G of first Betti number one and every d, by giving $\mu_{d,G}$ in closed form.

In general it is false. The counterexample is due to Chris Hall [12]: a bipartite graph on 41 vertices with $\mu_G = x^{21}(x^4 - 11x^2 + 25)^4(x^2 - 5)(x^2 - 11)$, where $\sqrt{5}$ lies in an internal gap of $\mathrm{spec}(T)$.

The natural repairs, minimum degree three and 3-connectivity, are open. Our contribution is to show that the obvious attack on them cannot work. A separator of size k met by p identical branches forces $A^{p-k} \mid \mu_G$, where A is the branch's matching polynomial (Lemma 7), so it produces roots of μ_G at will; but when the branch is a tree that same configuration is a θ-Aomoto subset in the sense of Banks, Garza-Vargas and Mukherjee, so the root it produces is an *eigenvalue* of the cover (Proposition 10). The condition $p > k$ that makes the divisor nontrivial is literally the Aomoto inequality $|\partial S| < \mathrm{cc}(G[S])$. The mechanism manufactures points of $\mathrm{spec}(T)$, not counterexamples.

This explains the one feature of Hall's family that had resisted explanation, the pendant leaves. They create the root $\sqrt{5}$ and simultaneously double the boundary of the relevant star, carrying it away from the Aomoto inequality; removing them destroys both. A counterexample must give its branches a private boundary, which is a condition on attachment rather than on degrees, and that is where a repair should be sought.

Everything asserted is machine-checked in Lean 4/Mathlib unless labelled otherwise. Supplementary material is in [18]; the biregular case is [17].

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Code and data availability. All computations, the Lean 4/Mathlib sources, and the scripts backing every numerical statement are in the `rama-notes` repository. Each numerical claim names the script that produced it; every named script is executed and its output compared against a recorded baseline by `code/cite_check.py`, and every Lean declaration cited is checked for its axiom base by `code/audit_papers.py`.

Status conventions. Claims here are of three kinds, and the text says which each time rather than leaving it to the reader. A result described as *formalized*, or cited by a Lean declaration name, is machine-checked in Lean 4/Mathlib: the sources are in `RamaLean/` of the accompanying repository, carry no `sorry`, and rest only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. A claim described as *measured*, *verified to* a stated tolerance, or *checked numerically* is computational evidence and is not a proof; the script producing it is named, and every named script is run and its output compared against a recorded baseline (`code/cite_check.py`). A claim called a *conjecture* is unproved, and is called that even where the evidence is strong.

1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [11], who used the $r = 2$ (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation r-lift* of a graph G replaces each vertex by r copies and each edge $\{u, v\}$ by the perfect matching $u_i \sim v_{\sigma(i)}$ of a permutation $\sigma \in S_r$ chosen independently and uniformly per edge. For the cycle C_n, gauge-fixing a spanning path reduces the lift to a single permutation $\sigma \in S_r$ on the closing edge, and the lift decomposes into disjoint cycles: each ℓ-cycle of σ contributes a cycle $C_{n\ell}$ (fixed points contribute copies of C_n). Since $\chi_{C_m}(x) = \det(xI - A_{C_m}) = 2T_m(x/2) - 2$, the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) \;=\; \mathbb{E}_{\sigma\in S_r} \prod_{\ell \,\in\, \mathrm{cycles}(\sigma)} \bigl(2T_{n\ell}(x/2) - 2\bigr). \tag{1}$$

Theorem 1. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) \;=\; U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) \;=\; \underbrace{\bigl(2T_n(x/2) - 2\bigr)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}\bigl(T_n(x/2)\bigr)}_{\Psi_r(x)}.$$

The factored form identifies the quotient Ψ_r, by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the d-matching polynomial ($d = r - 1$), as

$$\mu_{d,C_n}(x) \;=\; U_d\bigl(T_n(x/2)\bigr),$$

which is equivalent (through $\mathcal{P}_d(2t) = U_d(t)$, $\mathcal{C}_n(2t) = 2T_n(t)$ and the composition identity $U_{dn+n-1} = U_d(T_n)\,U_{n-1}$) to *Hall's conjecture* $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$, proved combinatorially in [3]. That preprint has remained unrefereed since 2018 and carries no journal reference or DOI as of August 2026. The reason is not mathematical. Chris Hall, who posed the conjecture, has told us (personal communication, August 2026) that Andrew Herring was his first doctoral student; Hall gave him the conjecture, Herring presented it as a problem at a workshop, and the group of [3] solved it there. They submitted the result, met a couple of rejections, and abandoned the attempt for want of motivation rather than for any defect. Hall regards the proof as correct, and so do we. We record the episode because a reader weighing what is established about this identity should know both that the combinatorial proof stands and that the present derivation is, as far as we can tell, the first refereed or machine-checked one. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

What this paper does

The closed form is the starting point rather than the subject. Read as a statement about where the roots of $\mu_{d,G}$ sit, it says that for the cycle they land exactly in $\mathrm{spec}(T)$, and not merely in the interval $[-\rho, \rho]$ that Hall–Puder–Sawin guarantee. The two differ whenever T has a spectral gap, and whether the stronger statement holds in general is the question here.

§3 proves it for every graph of first Betti number one and every d. §4 states it as a conjecture for all graphs. §5 is the counterexample, due to Chris Hall. §5.3 isolates the algebra behind it, and §5.4 shows that this algebra cannot be turned against the two natural repairs, because the configuration it needs is exactly an Aomoto subset. Supplementary material is in [18]; the biregular case, which survives the refutation, is [17].

2 Proof

Write $f(\ell) = 2T_\ell(Y) - 2$ with $Y = T_n(x/2)$; by the composition identity $T_{n\ell} = T_\ell \circ T_n$ this is the per-cycle factor in (1).

Step 1 (exponential formula). For any commutative ring R and $f : \mathbb{N} \to R$, the S_r-average $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \mathrm{cycles}(\sigma)} f(\ell)$ satisfies

$$r\,\Phi_r = \sum_{k=1}^{r} f(k)\,\Phi_{r-k}, \qquad \Phi_0 = 1, \tag{2}$$

the coefficient form of $\sum_r \Phi_r z^r = \exp\bigl(\sum_\ell f(\ell) z^\ell/\ell\bigr)$. (Proof: group σ by cycle type λ with Cauchy's count $r!/z_\lambda$, mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

Step 2 (Chebyshev generating function). With $f(\ell) = 2T_\ell(Y) - 2$,

$$\sum_{\ell \geq 1} f(\ell) \frac{z^\ell}{\ell} = -\log\bigl(1 - 2Yz + z^2\bigr) + 2\log(1 - z) \quad \Longrightarrow \quad \sum_{r \geq 0} \Phi_r z^r = \frac{(1-z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against $\sum U_r(Y) z^r = (1 - 2Yz + z^2)^{-1}$ gives $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$, and the three-term recurrence $U_r + U_{r-2} = 2YU_{r-1}$ collapses this to $(2Y - 2)\,U_{r-1}(Y)$. $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with $\Phi_0 = 1$ determines Φ over a characteristic-zero domain.

3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because C_n has $|E| - |V| + 1 = 1$. The argument therefore applies verbatim to any connected G with $b_1(G) = |E| - |V| + 1 = 1$, a cycle with trees attached, and we record what it gives.

By [2] the expected characteristic polynomial of a random r-lift of *any* graph factors as $\chi_G \cdot \mu_{r-1,G}$; the content below is therefore not that factorization but a closed form for the factor $\mu_{d,G}$ itself. Question 6.5 of [2] asks which parts of the theory of the matching polynomial $\mu_{1,G}$ generalize to $\mu_{d,G}$; the following answers it for $b_1(G) = 1$.

Let C be the unique cycle of G and let $G - V(C)$ be the forest obtained by deleting its vertices. Write μ_H for the ordinary matching polynomial of a graph H, with $\mu_\emptyset = 1$. Define

$$V_0 = 1, \qquad V_1 = \mu_G, \qquad V_d = \mu_G\, V_{d-1} - \mu^2_{G-V(C)}\, V_{d-2}. \tag{3}$$

Theorem 2. *For every connected G with $b_1(G) = 1$ and every $d \geq 0$, $\mu_{d,G} = V_d$.*

Theorem 2 is proved below, through the twisted characteristic polynomial; it is not formalized.

For $G = C_n$ the deleted forest is empty, so $\mu_{G-V(C)} = 1$ and (3) is the recurrence for $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$: Theorem 1 is the case $\mu_{G-V(C)} = 1$. In general the attached trees enter only through $\mu_{G-V(C)}$; for the triangle with one pendant edge, $\mu_G = x^4 - 4x^2 + 1$ and $\mu_{G-V(C)} = x$.

The proof goes through the twisted characteristic polynomial. Let $e = uv$ be the unique non-tree edge, let $A_G(z)$ be the adjacency matrix of G with the (u, v) entry replaced by z and the (v, u) entry by z^{-1}, and set $F(x, z) = \det(xI - A_G(z))$.

Lemma 3. *$F(x, z) = \mu_G(x) - \mu_{G-V(C)}(x)\,(z + z^{-1})$.*

Proof. By the Sachs coefficient theorem $\det(xI-A)$ expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and $A_G(z)^\mathsf{T} = A_G(z^{-1})$, the determinant is of the form $P(x) + Q(x)(z + z^{-1})$. Replacing z by ± 1 and averaging kills exactly the elementary subgraphs whose cycle traverses e; as G has the single cycle C, those are the subgraphs containing C, so $P = \mu_G$, and the surviving terms contribute $Q = -\mu_{G-V(C)}$, a cycle component carrying the Sachs weight -2. □

Proof of Theorem 2. Gauge-fix a spanning tree: the r-lift is determined by one permutation $\sigma \in S_r$ on e, and its components are indexed by the cycles of σ, the component of an ℓ-cycle being the ℓ-fold cyclic cover G_ℓ. Decomposing the $\mathbb{Z}/\ell$ action into characters gives the classical factorization $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$ with $\omega = e^{2\pi i/\ell}$ [9, 10]. With P, Q as in Lemma 3 and $y = -P/(2Q)$,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1}\Big(P + 2Q\cos\tfrac{2\pi j}{\ell}\Big) = (-Q)^\ell\big(2T_\ell(y) - 2\big),$$

so by the exponential formula (2), with $w = -Qz$,

$$\sum_{r\geq 0} \Phi_{G,r} z^r = \exp\Big(\sum_{\ell\geq 1}\big(2T_\ell(y) - 2\big)\frac{w^\ell}{\ell}\Big) = \frac{(1-w)^2}{1 - 2yw + w^2},$$

and extracting coefficients as in Section 2 gives $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1} U_{r-1}(y)$. Comparing with $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ [2] yields $\mu_{d,G} = (-Q)^d U_d(y)$, which satisfies (3). $\square$

Since $U_d(y) = 2^d \prod_{k=1}^{d}(y - \cos\frac{k\pi}{d+1})$, the theorem can be restated as a product, $\mu_{d,G}(x) = \prod_{k=1}^{d} F\big(x, e^{ik\pi/(d+1)}\big)$, whence:

This is the real-rootedness theorem of [2] for this family, with the roots identified. The localization itself, that a vanishing V_d forces $|\mu_G| \le 2|\mu_{G-V(C)}|$, so that a root can never sit in a spectral gap, is machine-checked in Lean 4 as `cV_root_mem_band`, resting on `cU_ne_zero_of_one_le_abs`: the Chebyshev polynomial U_d has no root of absolute value at least one.

The roots can be located much more precisely than that. Since $b_1(G) = 1$ the group $\pi_1(G)$ is $\mathbb{Z}$, so the universal cover T of G *is* the $\mathbb{Z}$-cover, and T is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres $A_G(e^{i\theta})$. Hence

$$\operatorname{spec}(T) = \bigcup_{\theta} \operatorname{spec} A_G(e^{i\theta}) = \Big\{x : \ \exists\, s \in [-1,1],\ \mu_G(x) = 2s\, \mu_{G-V(C)}(x)\Big\},$$

a finite union of bands, together with a flat band at each common zero of μ_G and $\mu_{G-V(C)}$, where $F(x,\cdot)$ vanishes identically.

Corollary 4. *For $b_1(G) = 1$ the roots of $\mu_{d,G}$ are exactly the solutions of $\mu_G(x) = 2\cos\big(\frac{k\pi}{d+1}\big)\mu_{G-V(C)}(x)$ for $1 \le k \le d$. In particular every root of every $\mu_{d,G}$ lies in* $\operatorname{spec}(T)$, *and as $d \to \infty$ the roots equidistribute on* $\operatorname{spec}(T)$ *with respect to its density of states.*

Proof. Immediate from Theorem 2 in the form $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$ and $F(x, e^{i\theta}) = \mu_G - 2\cos\theta\, \mu_{G-V(C)}$. The equidistribution is the arcsine law for $\cos\big(k\pi/(d+1)\big)$ transported through the fibre map. $\square$

Hall–Puder–Sawin place the roots of $\mu_{d,G}$ in the Ramanujan *interval* $[-\rho, \rho]$, ρ being the spectral radius of T. Corollary 4 places them in $\operatorname{spec}(T)$, which is a union of bands and is a proper subset of that interval whenever T has a spectral gap: no root of any $\mu_{d,G}$ can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where $\mu_{G-V(C)} = 1$ and $\operatorname{spec}(T) = [-2, 2]$ is the whole interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of $[-\rho, \rho]$ that is a gap):

G	ρ	gap proportion
C_n	2	0%
triangle + pendant	2.170	36.5%
C_4 + pendant	2.136	45.9%
triangle + P_2	2.214	51.3%
$C_4 + P_3$	2.189	54.1%
triangle + $K_{1,3}$	2.514	65.9%

So the cycle is not merely the easiest case of Theorem 2: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

The boundary is sharp. For $b_1(G) = b$ the lift is determined by b permutations, that is by an action of the free group F_b on $[r]$, whose orbits are the components of the lift. The exponential formula still applies, in the form $N_r = \sum_\ell \binom{r-1}{\ell-1} C_\ell N_{r-\ell}$ with $N_r = (r!)^b \Phi_{G,r}$ and C_ℓ the sum of χ over the transitive b-tuples on $[\ell]$; we have verified this for $b \leq 3$. What fails is the atom. For $b = 1$ every transitive tuple gives the *same* cover G_ℓ, so $C_\ell = (\ell-1)!\,\chi_{G_\ell}$: one graph per ℓ, with χ_{G_ℓ} Chebyshev in ℓ. For $b \geq 2$ the transitive tuples number $(\ell-1)!$ times the count of index-ℓ subgroups of F_b and spread over many non-isomorphic covers with distinct spectra, so C_ℓ carries no such structure. Accordingly Theorem 2 is not merely unproved for $b \geq 2$: it is false there already at $d = 1$, as we checked for the theta graph, the bowtie, K_4, $K_{2,3}$ and two triangles joined by an edge.

Theorem 2 was verified by exact integer computation against the definition of $\mu_{d,G}$ as an average of matching polynomials over all d-covers, and against brute-forced lift averages, on thirteen named unicyclic graphs for $r \leq 6$ and on a random sample of thirty-four for $r \leq 5$; Lemma 3 was checked on eleven.

4 A conjecture for all graphs

Corollary 4 is proved by Floquet theory, which is available only because $\pi_1(G) = \mathbb{Z}$. Its *statement*, however, makes sense for every graph, and we know one case of it: at $b_1(G) = 1$ it is Corollary 4, for every d. We therefore propose:

Conjecture 5. *For every finite graph G and every $d \geq 1$, all roots of $\mu_{d,G}$ lie in* $\mathrm{spec}(T)$, *the spectrum of the universal cover of G.*

Theorem 6 (Hall [12])**.** *Conjecture 5 is false, already at $d = 1$ and already for simple connected loopless bipartite graphs.*

The counterexample and its certificate are entirely due to Chris Hall. We record it here because it settles the central question of this paper, and because everything else in the note has to be read in its light. §5 states his construction, our independent verification of it, and what we can add.

The biregular case of Conjecture 5 at $d = 1$ is not new. It is Problem 1 of Song, Fan and Miao [5], posed in 2023: they ask whether the matching polynomial of the incidence graph B_H of a (d,r)-regular hypergraph has no root μ with $0 < |\mu| < |\sqrt{d-1} - \sqrt{r-1}|$, and every (d,r)-biregular bipartite graph is such an incidence graph. Their Theorem 1.8 attaches a consequence: an affirmative answer would give every (d,r)-regular hypergraph a left-sided Ramanujan 2-covering. The only bound in that direction we know of is their Lemma 5.5, which goes the other way, $0 < \mu_\tau(B_H) \leq \max\{\sqrt{d}, \sqrt{r}\}$. We are not aware of any proof, partial result or counterexample. Conjecture 5 extends their problem in two directions, to arbitrary graphs and to all d.

Question 6.3 of [2] asks the corresponding question for graph eigenvalues, and leaves it open: it distinguishes graphs whose nontrivial spectrum lies in the *interval* $[-\rho, \rho]$ from those whose nontrivial spectrum lies in $\mathrm{spec}(T)$, observes that the two notions separate exactly when T has a gap, and gives the biregular tree as the example. Conjecture 5 is the matching-polynomial form of that distinction.

The conjecture strengthens the root-localization theorem of [2], which places the roots in the interval $[-\rho, \rho]$ with ρ the spectral radius of T; the conjecture places them in $\mathrm{spec}(T)$

itself. The two agree whenever T has no spectral gap, in particular for regular G, where T is a regular tree and $\mathrm{spec}(T) = [-2\sqrt{q}, 2\sqrt{q}]$ is an interval. They differ exactly when T has a gap, and the smallest examples are biregular: for $K_{2,3}$ the universal cover is the $(3,2)$-biregular tree, with spectrum $\{0\} \cup \pm[\sqrt{2}-1, \sqrt{2}+1]$ and hence a gap $(0, \sqrt{2}-1)$ inside the Ramanujan interval. We have tested the conjecture on the biregular family, where the universal cover is an (a,b)-biregular tree with spectrum $\{0\} \cup \pm[\,|s-t|, s+t\,]$, $s = \sqrt{a-1}$, $t = \sqrt{b-1}$, so the gap is $(0, |s-t|)$. In each case $\mu_{d,G}$ was computed by averaging matching polynomials over all d-covers. No root ever lies in the gap; we also record the smallest nonzero $|\mathrm{root}|$ as a multiple of the gap edge, which measures how close the bound comes to being attained.

G	b_1	type	gap edge	d	$\min\|\mathrm{root}\|/\text{gap edge}$
$K_{2,3}$	2	$(3,2)$	0.4142	$1,2,3,4$	2.72, 2.19, 1.94, 1.78
$K_{2,4}$	3	$(4,2)$	0.7321	$1,2,3$	1.93, 1.65, 1.52
$K_{2,5}$	4	$(5,2)$	1.0000	$1,2,3$	1.66, 1.46, 1.37
$K_{3,4}$	6	$(4,3)$	0.3178	$1,2$	3.04, 2.31
$K_{3,5}$	8	$(5,3)$	0.5858	$1,2$	2.10, 1.73
subdivision of K_4	3	$(3,2)$	0.4142	$1,2$	1.97, 1.67

All of these have $b_1 \geq 2$, so none is covered by Corollary 4. It is worth recording that even $d = 1$ is not formal. There $\mu_{1,G}$ is the ordinary matching polynomial, whose roots are eigenvalues of the path tree of G (Godsil), and the path tree embeds in T; but eigenvalues of a finite subtree of T need not lie in $\mathrm{spec}(T)$ when T has a gap. For the $(3,2)$-biregular tree, whose gap is $(0, \sqrt{2}-1)$, the path P_8 is a subtree with eigenvalue $2\cos(4\pi/9) = 0.3473$ inside that gap. So the $d = 1$ case does not follow from the path-tree argument. It is nevertheless a theorem for an infinite family.

5 Hall's counterexample

Everything in this section is due to Chris Hall, who constructed the graph, found the certificate, and wrote the verification note [12]. Our contribution is the independent check, the formalization of the algebra. The two-parameter family containing his graph is in [18].

5.1 The graph and its matching polynomial

Let G consist of a central vertex c; for $1 \leq i \leq 5$ a copy of $K_{2,5}$ with degree-five vertices v_i, w_i and middle vertices $u_{i,1}, \ldots, u_{i,5}$; a pendant leaf ℓ_i at each w_i; and the edges cv_i. It is simple, connected, loopless and bipartite, with 41 vertices, 60 edges and $b_1(G) = 20$.

Writing H for one branch after deleting c, the vertex-deletion recurrence gives $\mu_{H-v} = x^5(x^2-6)$ and $\mu_H = x^4(x^4 - 11x^2 + 25)$, and then

$$\mu_G(x) = \mu_H(x)^4\big(x\mu_H(x) - 5\mu_{H-v}(x)\big) = x^{21}(x^4 - 11x^2 + 25)^4(x^2-5)(x^2-11), \qquad (4)$$

so $\sqrt{5}$ is a simple root. Identity (4), the simplicity of the root, and the value of the cofactor are machine-checked in `RamaLean/HallCounterexample.lean` as an identity in $\mathbb{Z}[X]$ (`muG_eq`, `muG_root_sqrt5`, `muG_root_simple`).

5.2 The spectral exclusion

Angel, Friedman and Hoory [13] characterise invertibility of $A_T - \lambda I$ by a finite nonzero *ratio system*: an assignment r_e to directed edges with $\lambda = 1/r_e + \sum_{e\to f} r_f$, whose decay matrix $\mathcal{R}_{e,f} = |r_f|^2$ on non-backtracking pairs has spectral radius below one. Hall exhibits one at $\lambda = \sqrt{5}$ with ratios in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$; its 120-state decay matrix has ten transient states at the leaves, and the recurrent block of 110 reduces by symmetry to a 6×6 orbit quotient K with $x = (20, 121, 47, 33, 28, 16)^\mathsf{T}$ satisfying $Kx < x$ componentwise, so $\rho(K) < 1$ by Perron–Frobenius. Hence $\sqrt{5} \notin \operatorname{spec}(A_T)$, which with (4) proves Theorem 6. We rebuilt every object from the graph rather than copying the displayed matrices (`code/hall_certificate.py`) and machine-checked the algebra (`RamaLean/HallCounterexample.lean`).

5.3 The Divisibility Lemma

The counterexamples are all instances of one algebraic fact, which we state once.

Lemma 7 (Divisibility)**.** *Let G be a finite graph, $S \subseteq V(G)$ with $|S| = k$, and suppose $G - S$ has p components, each inducing a copy of the same graph B. Write $A = \mu_B$. Then*

$$A^{p-k} \mid \mu_G.$$

Proof. Expand $\mu_G = \sum_M (-1)^{|M|} x^{n-2|M|}$ over matchings M of G and group the terms by

$$J(M) = \{\, i : \text{some edge of } M \text{ joins } S \text{ to the } i\text{-th component} \,\}.$$

A matching uses each vertex of S at most once, so distinct components in $J(M)$ are entered through distinct vertices of S; choosing one such vertex per component is an injection $J(M) \hookrightarrow S$, whence $|J(M)| \le k$. For $i \notin J(M)$ the restriction of M to the i-th component is an arbitrary matching of B, and summing over those choices independently contributes a factor $A^{p-|J(M)|}$. Each term of the resulting sum is therefore divisible by A^{p-k}, and hence so is μ_G. □

At $k = 2$ this is the divisor A^{p-2} of the two-cut identity. The exponent is positive exactly when $p > k$, which is why every construction below needs at least $k + 1$ branches. The counting step and the algebraic step are machine-checked separately (`RamaLean/DivisibilityLemma.lean`: `card_branches_le_sep`, `dvd_sum_of_pow_terms`, `matching_expansion_dvd`); the matching expansion itself is not, Mathlib carrying no matching polynomial.

5.4 Why the branch mechanism cannot produce a counterexample

The natural repairs are local hypotheses on G. Minimum degree two fails and bounded maximum degree fails, so one is led to:

Conjecture 8 (D3)**.** *Every finite graph of minimum degree at least three satisfies* $\operatorname{Zeros}(\mu_G) \subseteq \operatorname{spec}(T_G)$.

Conjecture 9 (C2)**.** *Every finite* 3*-connected graph satisfies* $\operatorname{Zeros}(\mu_G) \subseteq \operatorname{spec}(T_G)$.

Both are open. We record here why the construction of §5.3, which is the obvious way to attack them, cannot settle either: the configuration that makes Lemma 7 produce a root of μ_G is the same configuration that puts that root *inside* $\mathrm{spec}(T_G)$.

Banks, Garza-Vargas and Mukherjee [14] characterise the eigenvalues of the universal cover: θ is an eigenvalue of T_G if and only if G has a *θ-Aomoto subset*, that is a set $S \subseteq V(G)$ with $G[S]$ a forest, θ an eigenvalue of every component of $G[S]$, and

$$|\partial_G S| < \mathrm{cc}(G[S]).$$

Li, Magee, Sabri and Thomas [15] prove the corresponding statement for the maximal abelian cover, in the form that θ is an eigenvalue of G^{ab} if and only if θ is a zero of $\mu_{G\setminus\Gamma}$ for every 2-regular subgraph Γ, and Spier [16] proves that T_G and G^{ab} have the same eigenvalues.

Proposition 10 (the branch obstruction)**.** *Let $\theta \in \mathbb{R}$ and let $S \subseteq V(G)$ with $|S| = k$. If at least $k+1$ of the components of $G - S$ induce forests having θ as an eigenvalue, then θ is an eigenvalue of T_G, and in particular $\theta \in \mathrm{spec}(T_G)$.*

Proof. Let U be the union of $j \geq k+1$ such components. Then $G[U]$ is a disjoint union of forests, hence a forest, with $\mathrm{cc}(G[U]) = j$ and θ an eigenvalue of every component. Every vertex outside U adjacent to U lies in S, so $|\partial_G U| \leq k < j$. Thus U is a θ-Aomoto subset and [14] applies. □

The components need not be isomorphic to one another, and need not be all of $G - S$; only their number relative to $|S|$ matters. Taking them to be p copies of a single tree B with θ a root of μ_B recovers the case that concerns us:

The hypotheses of Proposition 10 are then exactly those of Lemma 7 together with $p > k$, which is what the lemma needs for the exponent $p-k$ to be positive. So whenever the branch mechanism forces a root of μ_G, it forces that root into $\mathrm{spec}(T_G)$. The two conditions are the same condition:

$$p > k \quad \text{is} \quad |\partial_G S| < \mathrm{cc}(G[S]).$$

A shared separator keeps the boundary at k while the number of components grows with p, and the Aomoto inequality is then automatic.

This is why Hall's construction is subtler than it looks, and it explains the role of the pendant leaves, which [18] records as essential without a mechanism. His branches are not trees, so the obstruction does not apply to them directly; the natural forest inside a branch is the star $K_{1,5}$ at w_i, whose boundary is the pair $\{v_i, \ell_i\}$. With five branches that gives $\mathrm{cc} = 5$ and $|\partial| = 10$, comfortably on the wrong side of the Aomoto inequality. Deleting the pendant leaves brings the boundary down to 5, still not below 5, and it also destroys the root: the leafless graph has $\mu = x^{16}(x^4 - 15x^2 + 45)(x^4 - 10x^2 + 20)^4$, with no factor $x^2 - 5$. The leaves therefore do two jobs at once, creating the root and doubling the boundary of the star, and both are needed (`code/aomoto_obstruction.py`). The general form is checked there on non-isomorphic branches as well, a hub joined to both ends of a P_3 and of a P_7, where $\sqrt{2}$ is an eigenvalue of each path and the two components already outnumber the single boundary vertex.

The moral for a repair is that it must forbid the *private* boundary structure that Hall uses. Every counterexample must avoid having a θ-Aomoto subset at its offending root, which is primarily a condition on how branches attach. The ambient degrees are not irrelevant, however: they enter through a counting bound, which we record next because it cuts the search space in half.

5.5 A degree bound on the point spectrum

Proposition 11 (degree spread). *Let G be a graph with minimum degree $\delta \geq 2$ and maximum degree Δ. If T_G has a nonzero eigenvalue, then*

$$\Delta \ > \ 2(\delta - 1), \qquad \textit{equivalently} \qquad \Delta \geq 2\delta - 1.$$

Proof. By [14] there is a θ-Aomoto subset S with $\theta \neq 0$. Put $s = |S|$, $c = \mathrm{cc}(G[S])$, $b = |\partial S|$, and let e be the number of edges of G with exactly one end in S. Since $G[S]$ is a forest with c components it has $s - c$ edges, so counting the degrees of the vertices of S gives $e = \sum_{v \in S} \deg v - 2(s - c) \geq (\delta - 2)s + 2c$. Every such edge has its other end in ∂S, and a boundary vertex meets at most Δ edges in total, so $e \leq \Delta b$. As $\theta \neq 0$, no component of $G[S]$ is a single vertex, whence $s \geq 2c$; and the Aomoto inequality is $b \leq c - 1$. Combining,

$$2c(\delta - 1) \ = \ (\delta - 2)(2c) + 2c \ \leq \ (\delta - 2)s + 2c \ \leq \ e \ \leq \ \Delta b \ \leq \ \Delta(c - 1) \ < \ \Delta c,$$

and dividing by $c > 0$ gives $2(\delta - 1) < \Delta$. □

Taking $\Delta = \delta$ recovers the classical fact that a regular tree has no eigenvalues. The case that matters here is the contrapositive.

Corollary 12. *If $\delta(G) \geq 3$ and $\Delta(G) \leq 2\delta(G) - 2$, then T_G has no nonzero eigenvalue. In particular this holds for every graph with degrees in $\{3, 4\}$.*

The counting step and both consequences are machine-checked in `RamaLean/DegreeBound.lean` as `degree_bound`, `no_eigenvalue_of_small_spread` and `mindeg_three_maxdeg_four`; as in Proposition 10, the criterion of [14] is the analytic input and is cited rather than reproved, and the three combinatorial facts above enter the Lean statement as hypotheses.

Corollary 12 is what makes the bound useful rather than decorative. A counterexample to Conjecture 8 needs a root of μ_G that is outside the bands of $\mathrm{spec}(T_G)$ *and* not an eigenvalue, and it was the second condition that defeated every attempt in [18]: each time, the root that appeared to escape was carried back into the spectrum by an Aomoto subset. In the range $\Delta \leq 2\delta - 2$ that failure mode is impossible, so only the first condition remains. Every known counterexample to Conjecture 5, Hall's included, has $\delta = 1$, for which the bound is vacuous; this is consistent with the role his pendant leaves play above.

5.6 The origin: Hall's condition exactly

Proposition 11 needs $\theta \neq 0$, because the step $s \geq 2c$ used that no component of $G[S]$ is a single vertex. The excluded case is not a loose end. It has a complete and entirely elementary answer, which we record because it identifies the point spectrum at the origin with a classical condition.

Theorem 13. *For any graph G, the following are equivalent.*

1. 0 is an eigenvalue of the universal cover T_G.

2. G contains an independent set S with $|N(S)| < |S|$, that is, G fails Hall's condition.

Proof. By [14], (i) holds iff G has a 0-Aomoto subset.

If S is independent then every component of $G[S]$ is a single vertex, whose only eigenvalue is 0, and $\mathrm{cc}(G[S]) = |S|$ while $\partial_G S = N(S)$. So the Aomoto condition $|\partial_G S| < \mathrm{cc}(G[S])$ reads $|N(S)| < |S|$, and (ii) implies (i).

Conversely let S be a 0-Aomoto subset, with $G[S]$ a forest of components $T_1, \dots, T_c$ and $b := |\partial_G S| \le c - 1$. A forest has 0 as an eigenvalue iff it has no perfect matching, so each T_i has deficiency $d_i := |T_i| - 2\nu_i \ge 1$, where ν_i is its matching number. Choose a maximum independent set $S'_i \subseteq T_i$; as T_i is bipartite, König's theorem gives $|S'_i| = |T_i| - \nu_i = \nu_i + d_i$, and $N_{T_i}(S'_i) \subseteq T_i \setminus S'_i$ has at most ν_i elements. Put $S' = \bigcup_i S'_i$. Distinct components of an induced subgraph have no edges between them, so S' is independent in G, and every neighbour of S' lies in some $T_i \setminus S'_i$ or in $\partial_G S$. Hence

$$|S'| = \sum_i (\nu_i + d_i) \ \ge\ \sum_i \nu_i + c, \qquad |N(S')| \ \le\ \sum_i \nu_i + b \ \le\ \sum_i \nu_i + c - 1,$$

so $|N(S')| < |S'|$ and (ii) holds. □

The counting step is machine-checked in `RamaLean/HallZero.lean` as `hall_zero_counting` and `hall_violator_of_aomoto_zero`; as in Proposition 11, the criterion of [14] is the analytic input and is cited rather than reproved, and König's theorem and the deficiency of a forest without a perfect matching enter as hypotheses. The equivalence itself, and every step of the construction above, was checked by exhaustive enumeration of all 27474 connected graphs on at most six vertices, with 2765 positive instances and no discrepancy (`code/hall_zero.py`).

Two consequences are worth stating. First, an independent set violating Hall's condition admits no matching saturating it, so Theorem 13 gives at once that $0 \in \mathrm{spec}_p(T_G)$ implies G has no perfect matching, hence $\mu_G(0) = 0$. The converse fails: C_5 has no perfect matching and no Hall violator, and indeed its cover is the infinite path, whose spectrum $[-2, 2]$ contains 0 but not as an eigenvalue. Second, taken with Proposition 11, the point spectrum of a universal cover is now bounded by combinatorics from both sides: at the origin it is exactly a Hall violation, and away from the origin it requires the degree spread $\Delta > 2(\delta - 1)$.

5.7 An exact search in the trap-free class

Corollary 12 identifies a class in which a counterexample would be unconditional, which makes it worth searching directly. Among trap-free graphs the $(3, 4)$-biregular ones are the natural target, for a reason that is not obvious in advance: a random graph with degrees in $\{3, 4\}$ has no spectral gap at all, its cover spectrum running continuously through the origin, so there is nowhere for a root to hide. The biregular case is the exception. There

$$\mathrm{spec}(T) = \{0\} \cup \pm\big[\sqrt{3} - \sqrt{2},\ \sqrt{3} + \sqrt{2}\big] = \{0\} \cup \pm[0.317837\ldots,\ 3.146264\ldots],$$

and the complement has two windows. Both must be checked: Heilmann–Lieb confines the zeros of μ_G only to $|x| < 2\sqrt{3} = 3.4641\ldots$, which exceeds $\sqrt{3} + \sqrt{2}$, so the upper window is not empty on those grounds.

The search is exact, with no floating point in the verdict. Writing $\mu_G = x^{n-2\nu}Q(x^2)$, the polynomial Q is hyperbolic with all roots real and strictly positive, so its Budan–Fourier count is exact rather than an upper bound with even defect, and it requires no polynomial remainder sequence and hence suffers none of the coefficient growth of a Sturm chain. Matching counts are computed by a subset dynamic program over the smaller side of the bipartition and Q is evaluated in exact integer arithmetic at rational endpoints chosen to lie strictly inside the window being tested. Each graph additionally carries the free consistency check $V(0) = \deg Q$, and a graph failing it is counted as undecided rather than passed.

The sweep (`code/rust_biregular`) tested 82896 graphs: exhaustively at $t = 1$ and $t = 2$, meaning all 2895 connected labelled $(3, 4)$-biregular graphs on 14 vertices, and 20000 random graphs at each of $t = 3, \ldots, 6$, up to $n = 42$. No zero of μ_G lies outside $\mathrm{spec}(T_G)$ anywhere in the sample, and no graph failed the consistency check.

We state plainly what this is and is not. The individual verdicts are exact; the coverage is not exhaustive beyond $n = 14$, the random graphs are drawn by stub pairing with rejection and so are not uniform, and the run terminated because the range of t was exhausted rather than because a time budget expired. It is evidence for Conjecture 5 in the trap-free class, not a proof. Its value is that it is a negative result obtained where a positive one would have been unconditional: the one region of the class with a spectral gap, and the region in which the Aomoto escape is machine-checked to be unavailable, contains no counterexample up to 42 vertices.

5.8 The width threshold, and what it costs

The searches reported here and in [18] filter the detected gaps of $\mathrm{spec}(T)$ by a minimum width before looking for roots inside them, and skip a graph when no gap survives. The threshold used throughout is 0.05. Hall's own counterexample does not survive it: $\mathrm{spec}(T)$ for his graph has a gap

$$(2.220,\ 2.260), \qquad \text{width } 0.040, \qquad \sqrt{5} = 2.23607\ldots \in (2.220, 2.260),$$

and $0.040 < 0.05$, so his graph is scored as having no gap and reported clean (`code/gapfilter_blindspot.py`). The detector finds the gap; the threshold discards it.

Two measurements fix how much this costs, and they point in opposite directions.

Re-running the minimum-degree-three families at 0.015 changes nothing. Exactly 87 of the graphs swept present a gap at 0.05 and exactly 87 at 0.015, and no root in any of them is a violation (`code/rerun_lowgap.py`). For those families the threshold was not discarding search space, and the recorded verdicts stand unweakened. We had expected otherwise and record the miss.

But the threshold does exclude the width regime in which counterexamples actually occur. A search of one-cut constructions run at 0.015, with the exact eigenvalue test of §5.4 in place of a numerical one, returns four counterexamples to Conjecture 5 on 36 and 37 vertices, with minimal polynomials $x^6 - 12x^4 + 25x^2 - 7$, $x^6 - 13x^4 + 26x^2 - 7$, $x^6 - 13x^4 + 32x^2 - 13$ and $x^6 - 13x^4 + 32x^2 - 12$, decays 0.94 to 0.98, and gap widths of 0.040 in every case (`data/lowgap_counterexamples.json`). Together with Hall's, that is five known counterexamples, all of them in gaps of width 0.040, which is the smallest a scan of step 0.02 can resolve, and all of them below the threshold every search here used.

How wide are those gaps really? The scan reports 0.040 in all five cases, which is two of its 0.02 steps and therefore its resolution floor rather than a measurement. Resolving them needs a detector that returns contiguous intervals, which `code/gapscale2.py` supplies: it replaces the real fixed-point iteration by the complex cavity recursion at $z = \lambda + i\eta$, which converges at every λ, and separates band from gap from atom by the scaling of $|\operatorname{Im}\sum_w G_w|$ between two values of η a factor 100 apart. It reproduces the exact biregular gap $(0, |s-t|)$ for $K_{3,4}$ and $K_{2,3}$ and reports no gap for a 3-regular graph. Scanned at step 0.002, the gaps holding the five violating roots have widths

$$0.030, \qquad 0.002, \qquad 0.018, \qquad 0.026, \qquad 0.038,$$

every one below the 0.05 threshold that discarded them. The phenomenon really does live in gaps narrower than the filter, and the filter really does exclude every example known to us.

Two things we got wrong on the way to that are worth recording, since both are easy traps. The non-convergence of the real iteration is not a defect: it corresponds to λ lying in a band, and scoring it as spectrum is correct. And a fine scan that takes the extremes of the certified points merges disjoint gaps across an intervening band; doing that produced an apparent gap of width 0.1035 where there are really two gaps of 0.04 and 0.025 with a band between them (`code/gapwidth_underreport.py`, retracted in place).

The remedy for a search is therefore a finer step and a much lower cutoff, not a different solver. At step 0.02 a gap of width 0.002 cannot be seen at all.

The violating root sits at 40 to 61 percent of the way across its gap in four of the five cases and at 99 percent in the fifth, which is accordingly the least robust of the set.

One of the four, the 36-vertex example, carries a certified band exclusion as well. Its automorphism group has order 5040 and its 84 directed edges fall into 12 orbits of size 7, so the Angel–Friedman–Hoory system has twelve unknowns; solved at 60 digits it gives a Collatz–Wielandt constant $c = 0.9709248\ldots$ against a solution uncertainty of 5×10^{-58}, a margin of 55 orders of magnitude (`code/exact_certificate36.py`). That is short of Hall's certificate, which is exact in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$, and the shortfall is not removable by more effort: two of the twelve ratios lie in $\mathbb{Q}(\theta)$ with closed forms, and the other ten admit no polynomial relation of degree at most 30 with coefficients below 10^{20} at 200 digits. A large automorphism group does not give a small ratio field. We record also that integer relation search returned plausible expressions in $\mathbb{Q}(\theta)$ for all twelve, of which ten were spurious and were caught only by exact reduction in $\mathbb{Q}[x]/(f)$.

The four new examples have minimum degree one, so they bear on Conjecture 5 and not on Conjectures 8 or 9, which remain open. Re-running the minimum-degree-three families at the lower threshold changes nothing: 87 graphs present a gap at either setting and no root in any of them is a violation (`code/rerun_lowgap.py`), so those recorded verdicts stand unweakened.

5.9 Wheels saturate the bound

Wheels are the one family of minimum degree three here that carries genuine spectral gaps, so they are where a counterexample to Conjecture 8 would most plausibly appear. Re-checking them with the corrected detector of §5.8 and the exact eigenvalue test, and extending the range, none of $W_5, \ldots, W_{50}$ has a root of μ inside a gap. The roots straddle the gap: the bulk lie below its lower edge, near 2, and the largest lies above its upper edge.

What is new is the rate at which they do so. Writing

$$\mathrm{margin}(m) = \big(\text{largest root of } \mu_{W_m}\big) - \big(\text{top of the gap below it}\big),$$

the margin falls monotonically from 0.637 at $m = 5$ to 0.194 at $m = 50$, and a log-log fit over that range gives

$$\mathrm{margin}(m) \;\approx\; 1.463\, m^{-0.5125}, \qquad R^2 = 0.999,$$

so the family saturates the covering bound asymptotically without violating it (`code/wheel_saturation.py`). The computation uses the closed form $\mu_{W_m} = x(\mu_{P_m} - \mu_{P_{m-2}}) - m\,\mu_{P_{m-1}}$, checked against the deletion recursion before use.

This is the same phenomenon the companion paper records for the biregular family, where the margin decays like $n^{-2/3}$, the soft-edge exponent [17]. That figure has been replicated here independently, with matchings recounted by a subset dynamic program on independently generated graphs: the fit gives $n^{-0.6629}$ at $R^2 = 0.9996$, reproducing the published series to four decimals at $n = 12$ (`code/softedge_audit.py`). It is worth noting that the biregular claim needs no spectral instrument at all, the band edge for a biregular cover being the closed form $|\sqrt{q-1} - \sqrt{d-1}|$, so none of the failures recorded in §5.8 can touch it.

So two families saturate, at -0.51 and -0.66, and the exponents are genuinely different. We do not read that as two edge universality classes. The biregular family has bounded degree, d and q fixed as n grows, and the wheel family does not, since the hub of W_m has degree m. Under the soft-edge heuristic a margin $N^{-1/(1+\alpha)}$ corresponds to a density vanishing like $(x-\tau)^{\alpha}$, so $-2/3$ is the square-root edge $\alpha = \frac{1}{2}$ and $-\frac{1}{2}$ would be a linear edge $\alpha = 1$; but an unbounded degree could produce the same shift for unrelated reasons, and nothing here separates the two explanations.

What is unambiguous is the part that bears on Conjecture 8: both families saturate, the margin tends to zero in each, and an argument yielding a margin bounded below independently of size cannot prove the conjecture on either.

5.10 What this leaves

Conjecture 5 is false, by Hall's counterexample. Conjectures 8 and 9 are open, and Proposition 10 says that the branch mechanism cannot settle them in the negative: it manufactures eigenvalues of the cover, not counterexamples.

That is a sharper statement than it may look. Any counterexample must have its offending root avoid every θ-Aomoto subset, so a search should be organised around the Aomoto inequality rather than around degrees or connectivity. Hall's graph satisfies this by giving each branch a private boundary of two vertices; we know of no systematic way to do that at minimum degree three.

The biregular case, which survives the refutation of Conjecture 5 and is where the phenomenon lives, is [17].

6 Related work and formalization

The d-matching polynomial and the interval $[-\rho, \rho]$ are due to Hall, Puder and Sawin [2], whose Question 6.3 asks for exactly the strengthening considered here; Heilmann and Lieb [6] give the classical interval, Godsil [7, 8] the path-tree method, and Angel, Friedman and

Hoory [13] the ratio criterion the certificates use. The cycle identity behind Theorem 1 was proved combinatorially by Cochran, Groothuis, Herring, Rohatgi and Stucky [3]; our route is independent and elementary. Song, Fan and Miao [5] pose the biregular case, which is [17].

The Lean 4/Mathlib sources are in `RamaLean/` of the accompanying repository and carry no `sorry` and no custom axioms: the closed form at first Betti number one, the counterexample's algebra, and both halves of Lemma 7. The standing gap is that Mathlib has no matching polynomial, so the expansion in Lemma 7 and the Angel–Friedman–Hoory criterion are cited rather than derived.

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